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SUB CODE: DSA0502

SUB NAME: QUERY PROCESSING FOR DATA SCIENCE

CO4 ASSESSMENT TOOL 3

22.08.2026

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1. Correlation Analysis of Student Performance : Import student datasets containing attendance, assignment marks, internal marks, study hours, and final marks. Explore the tables, join relevant datasets, calculate correlations between variables, identify outliers, and use appropriate charts to explain the major patterns in student performance.

AIM: To analyze student performance by integrating datasets containing attendance, assignment marks, internal marks, study hours, and final marks, calculate correlations between different performance variables, identify outliers, and use appropriate charts to understand major patterns in student performance.

TOOLS USED: Python ,Pandas,NumPy, Matplotlib ,Seaborn ,VS Code.

DATASET USED: All datasets were connected using the common Student_ID field.

- Final Marks Dataset – Student ID and final examination marks.
- Students Dataset – Student ID, name, department, and semester.
- Attendance Dataset – Student ID and attendance percentage.
- Assignments Dataset – Student ID and assignment marks.
- Internal Marks Dataset – Student ID and internal marks.
- Study Hours Dataset – Student ID and study hours.

DATASET JOINING:

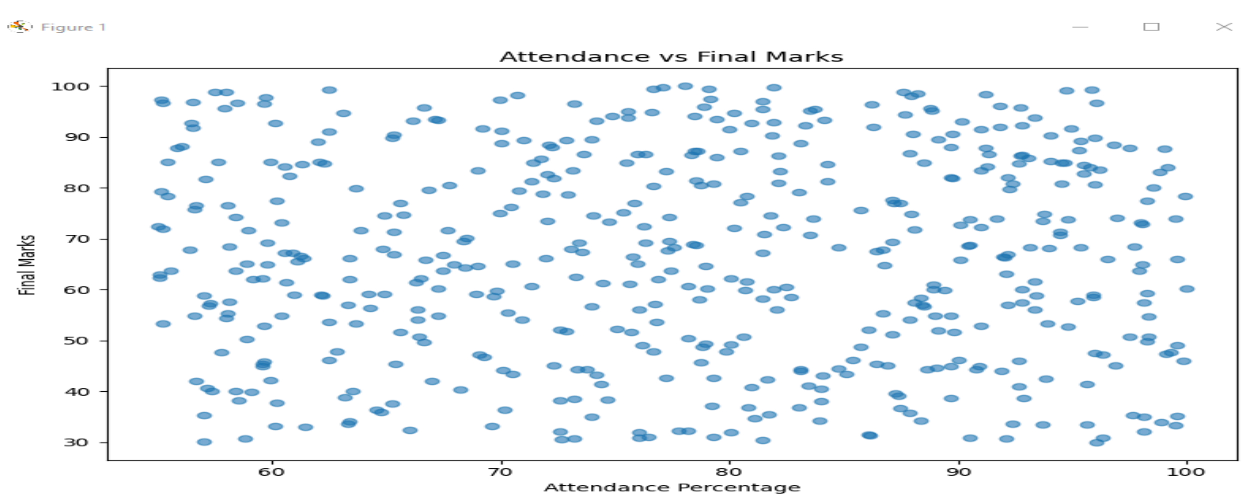
The datasets were integrated using the common Student_ID column.

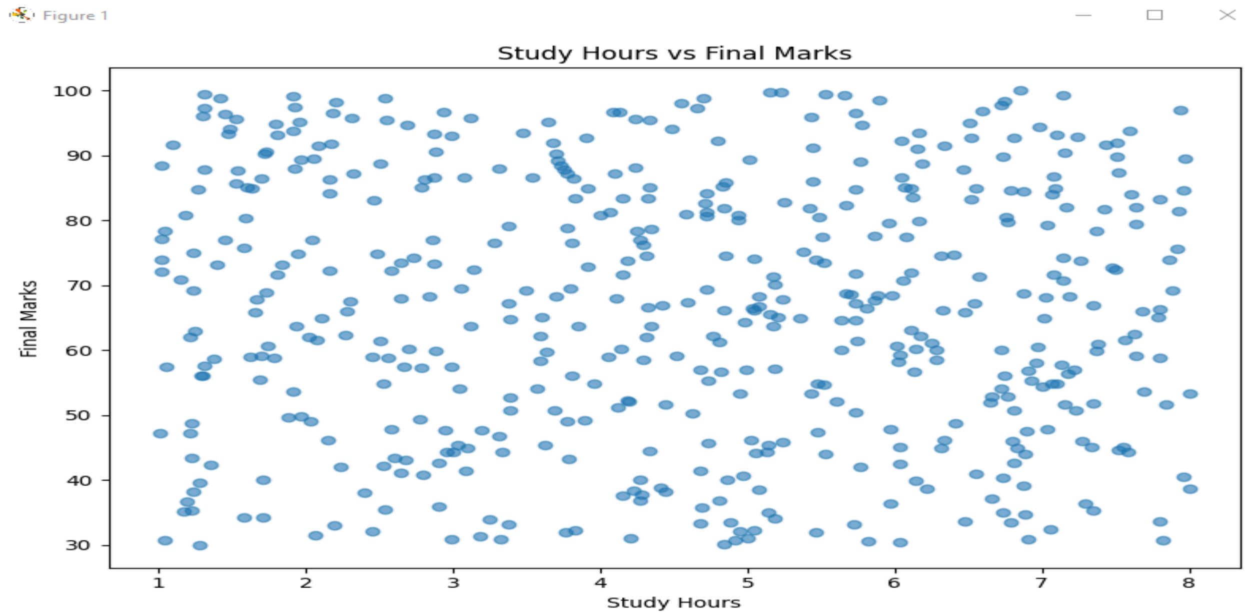
The following tables were joined: Students + Attendance + Assignments + Internal Marks + Study Hours + Final Marks

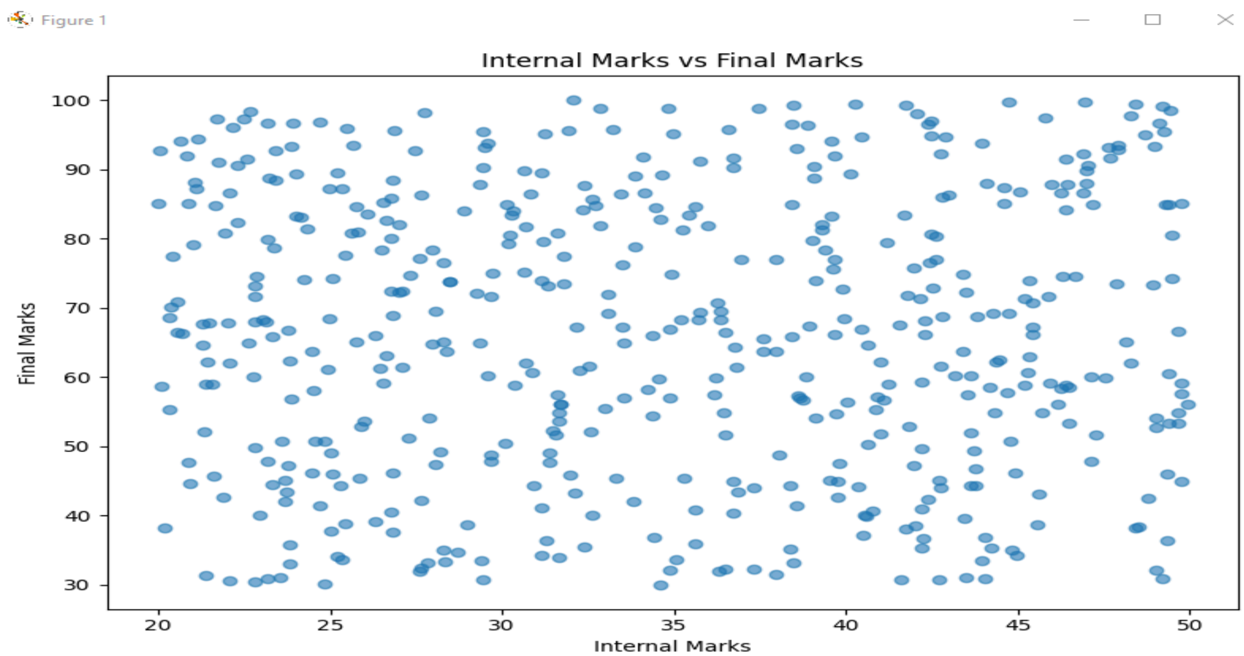
The resulting combined dataset contained information about each student's academic and study-related factors.

```
========== JOINED DATASET ==========
Shape: (500, 9)

Columns:
['Student_ID', 'Name', 'Department', 'Semester', 'Attendance_
Percentage', 'Assignment_Marks', 'Internal_Marks', 'Study_Hou
rs', 'Final_Marks']

First 5 rows:
   Student_ID       Name  ... Study_Hours  Final_Marks
0           1  Student_1  ...        4.97        40.66
1           2  Student_2  ...        2.99        92.96
2           3  Student_3  ...        6.77        79.71
3           4  Student_4  ...        1.52        95.62
4           5  Student_5  ...        2.58        72.23

[5 rows x 9 columns]
```

DATA VISUALIZATION:





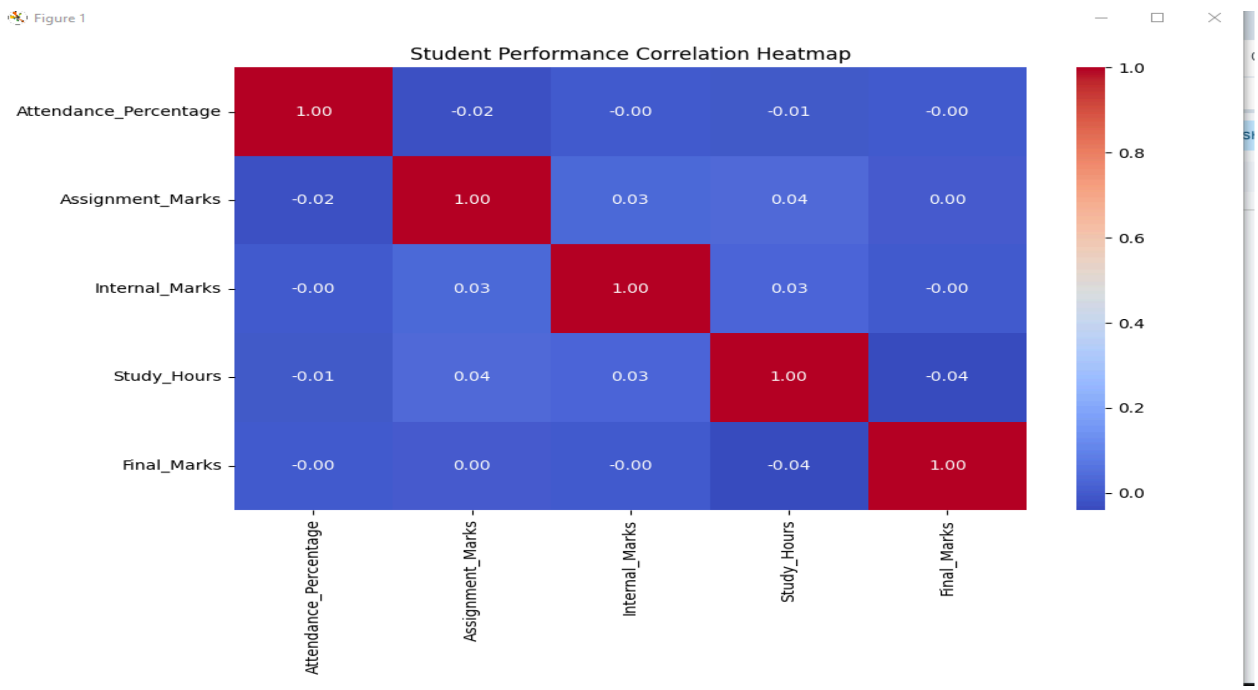


CORRELATION ANALYSIS:

Correlation analysis was performed to study the relationships between: Attendance Percentage,Assignment Marks ,Internal Marks,Study Hours,Final Marks

The correlation coefficient was used to determine the strength and direction of relationships.

A correlation value close to **+1** indicates a strong positive relationship, while a value close to **-1** indicates a strong negative relationship. A value close to **0** indicates a weak or no linear relationship.



OUTLIER DETECTION:

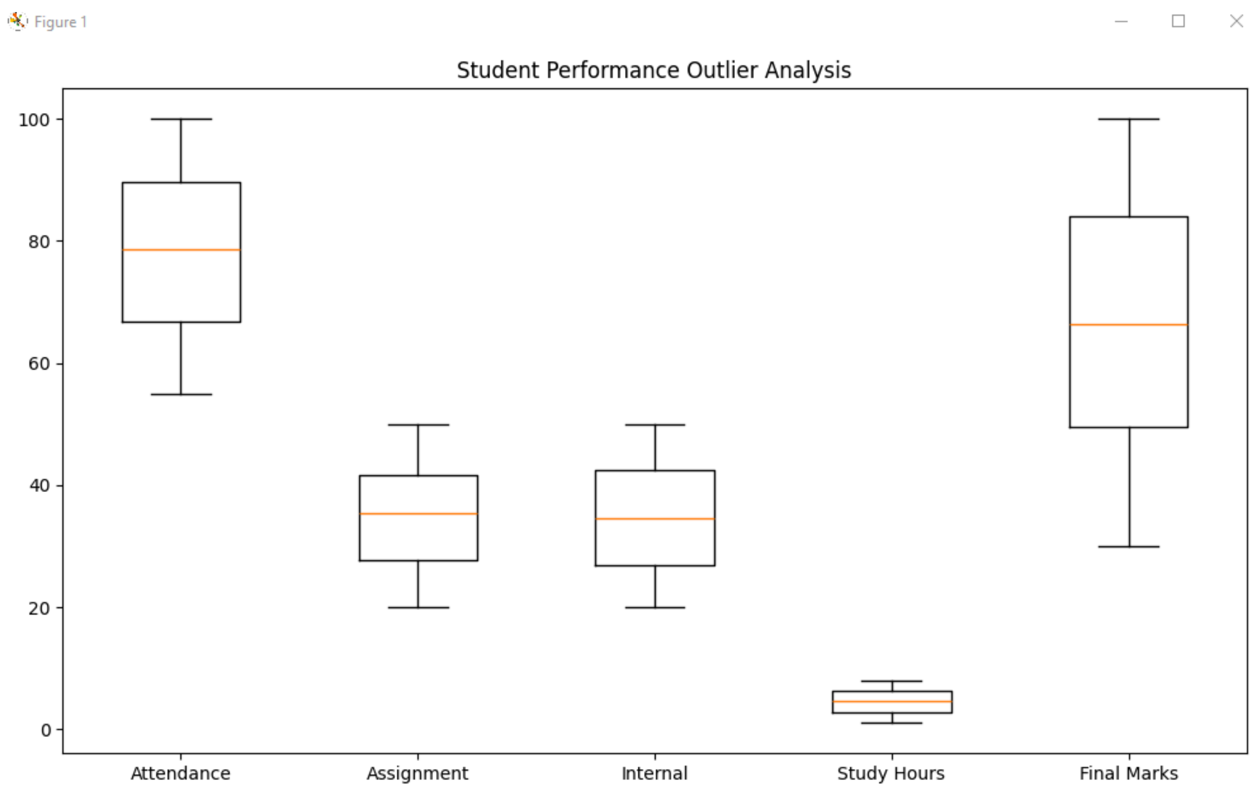
The **Interquartile Range (IQR) method** was used to identify potential outliers in:

- Attendance
- Assignment Marks
- Internal Marks
- Study Hours
- Final Marks

The IQR method calculates the difference between the third quartile (Q3) and first quartile (Q1).

IQR = Q3 − Q1

Values outside the calculated lower and upper limits were considered potential outliers.



KEY FINDINGS:

1. STRONGEST FACTOR

Study_Hours has the strongest correlation with Final Marks: -0.04.

2. AVERAGE FINAL MARKS

The average final examination mark is 66.10.

3. AVERAGE INTERNAL MARKS

The average internal mark is 34.72.

4. AVERAGE ATTENDANCE

The average student attendance is 78.11%.

5. OUTLIER ANALYSIS

A total of 0 potential outlier observations were identified across the analyzed variables.

CONCLUSION:

The student performance datasets were successfully integrated using the common Student_ID field. Correlation analysis was performed to understand the relationships between attendance, assignment marks, internal marks, study hours, and final examination marks.

The IQR method was used to identify potential outliers, while scatter plots, a correlation heatmap, and box plots were used to visualize the major patterns in student performance.

Overall, the analysis demonstrates how data integration, correlation analysis, outlier detection, and visualization can be used to understand student academic performance and support data-driven educational decision-making.

2.Sales and Customer Pattern Analysis : Import sales and customer datasets and combine them using suitable table-joining techniques. Analyze the correlation between customer age, purchase frequency, discount, and purchase amount. Create meaningful customer groups, identify unusual purchasing patterns, and present the results using suitable visualizations.

AIM: To analyze sales and customer data by combining relevant datasets, studying the relationships between customer age, purchase frequency, discount, and purchase amount, creating meaningful customer groups, identifying unusual purchasing patterns, and presenting the results using appropriate visualizations.

TOOLS USED: Python ,Pandas,NumPy, Matplotlib ,Seaborn ,VS Code.

DATASETS USED: customers.csv , sales.csv.

DATA EXPLORATION:

The customer and sales datasets were loaded and explored using Pandas. Dataset dimensions, column names, data types, missing values, and statistical summaries were examined before performing the analysis .

```
========== CUSTOMER STATISTICS ==========
       Customer_ID        Age
count   500.000000 500.000000
mean    250.500000  41.278000
std     144.481833  13.389072
min       1.000000  18.000000
25%     125.750000  30.000000
50%     250.500000  42.000000
75%     375.250000  52.000000
max     500.000000  64.000000

========== SALES STATISTICS ==========
           Sale_ID  Customer_ID  ...     Discount  Purchase_Amount
count  1000.000000  1000.000000  ...  1000.000000      1000.000000
mean    500.500000   259.303000  ...    20.328260      5152.283120
std     288.819436   146.899383  ...    11.366381      2757.567624
min       1.000000     1.000000  ...     0.080000       501.770000
25%     250.750000   133.000000  ...    10.757500      2749.257500
50%     500.500000   270.000000  ...    20.715000      5073.115000
75%     750.250000   382.250000  ...    30.117500      7569.952500
max    1000.000000   500.000000  ...    39.980000      9977.420000
```

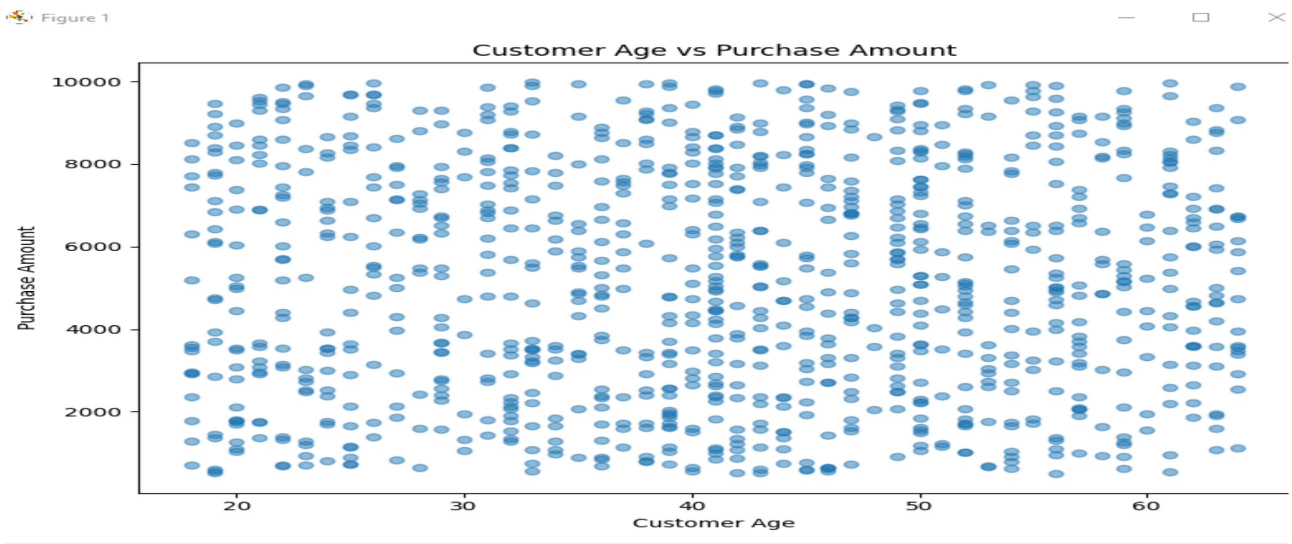
DATASET JOINING:

The customer and sales datasets were combined using an inner join on `Customer_ID`. This created an integrated dataset containing both customer demographic information and purchasing information.

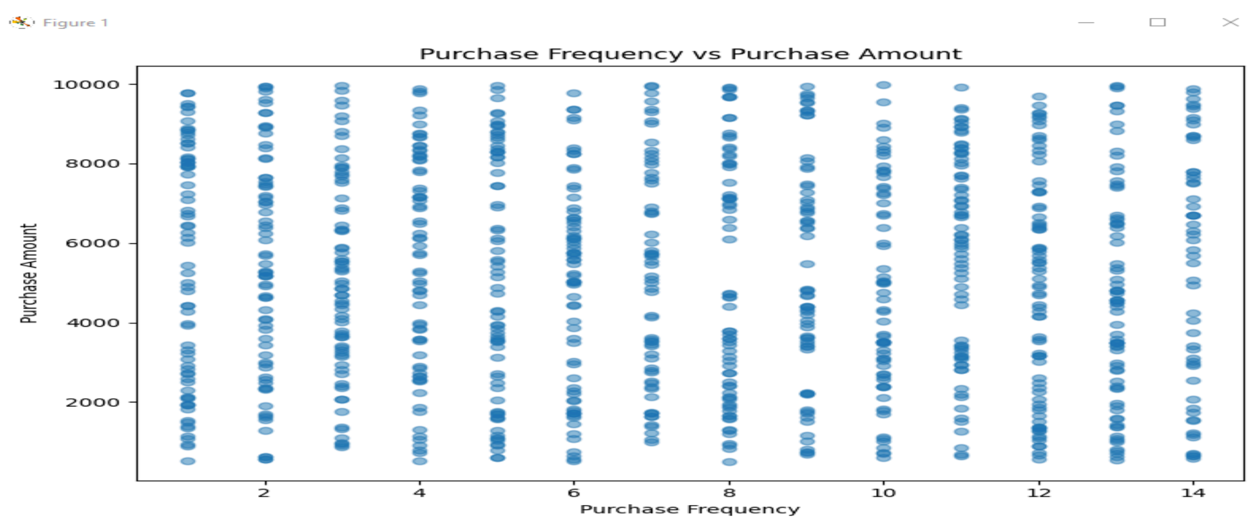
```
[8 rows x 5 columns]

========== JOINED DATASET ==========
Shape: (1000, 10)

Columns:
['Customer_ID', 'Customer_Name', 'Age', 'Gender', 'City', 'Sale_ID', 'Prod
uct_Category', 'Purchase_Frequency', 'Discount', 'Purchase_Amount']

First 5 Rows:
   Customer_ID Customer_Name  ...  Discount Purchase_Amount
0            1    Customer_1  ...      9.48         1300.43
1            1    Customer_1  ...      8.62         9260.82
2            1    Customer_1  ...     30.30         7521.37
3            2    Customer_2  ...     33.97         3645.89
4            2    Customer_2  ...      4.68         8919.74

[5 rows x 10 columns]
```

DATA VISUALIZATION:

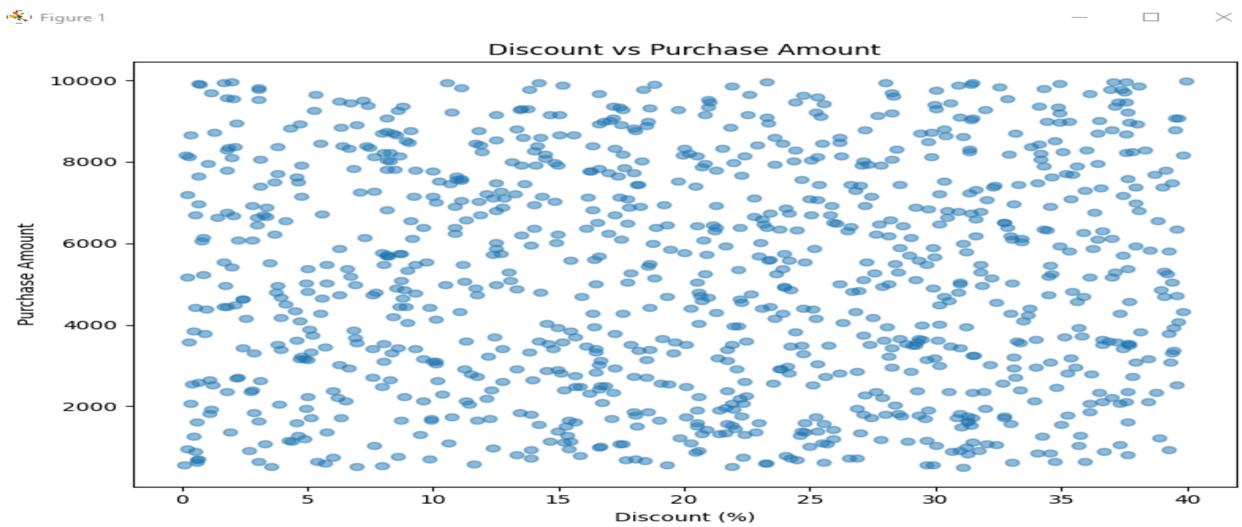
Customer Age vs Purchase Amount



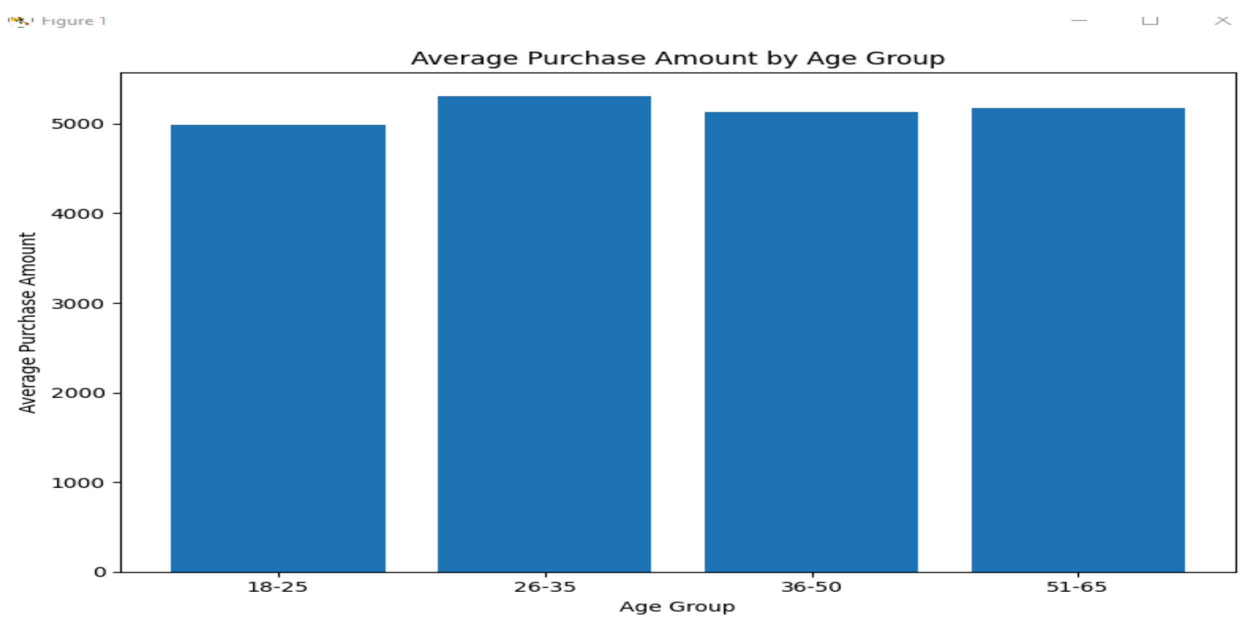
Purchase Frequency vs Purchase Amount



Discount vs Purchase Amount



Average Purchase Amount by Age Group

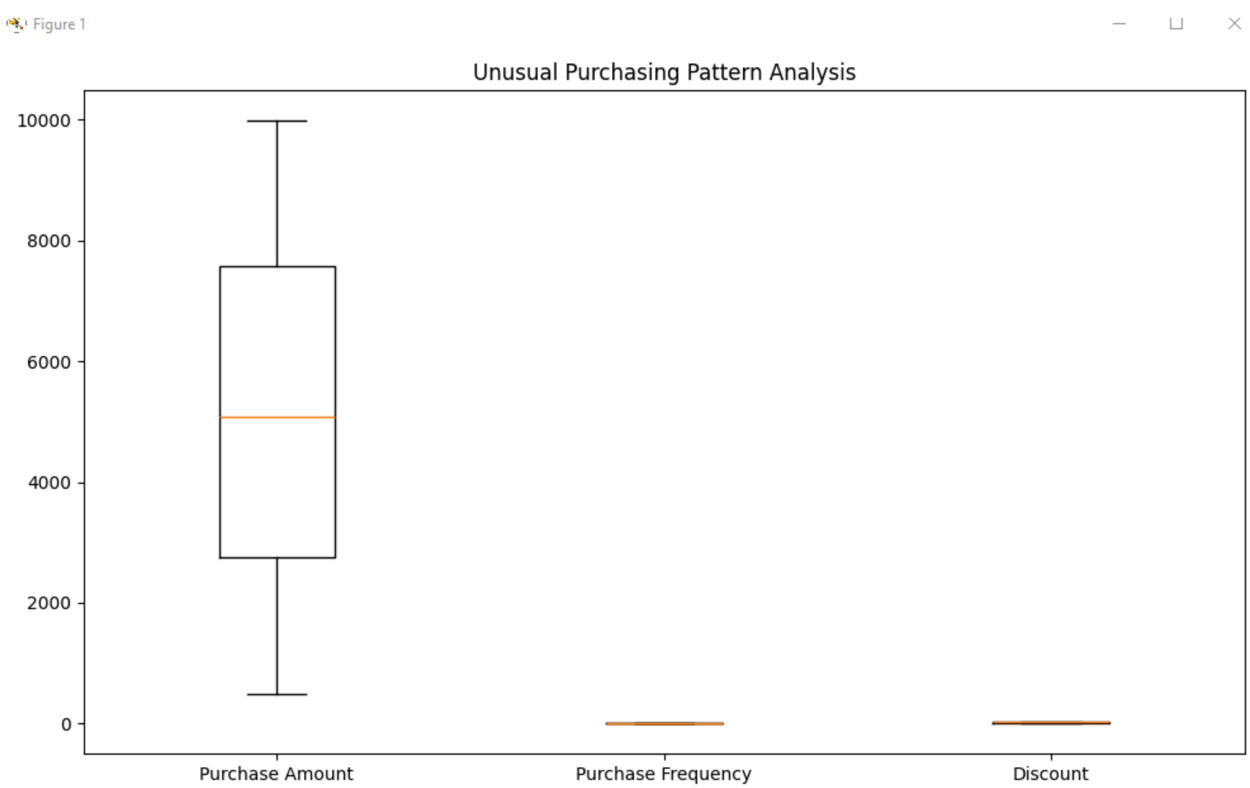


CORRELATION:

The correlation matrix was used to identify the strength and direction of relationships between these variables.

OUTLIER DETECTION:

The **IQR method** was used to identify unusual purchasing patterns. Potential outliers were analyzed for purchase amount, purchase frequency, and discount.



KEY FINDINGS:

1. STRONGEST CUSTOMER FACTOR

Purchase_Frequency has the strongest correlation with Purchase Amount: -0.04.

2. BEST CUSTOMER AGE GROUP

Age group 26-35 has the highest average purchase amount of 5304.57.

3. AVERAGE PURCHASE AMOUNT

The overall average purchase amount is 5152.28.

4. AVERAGE DISCOUNT

The average discount provided is 20.33%.

5. UNUSUAL PURCHASING PATTERNS

0 potential purchase amount outliers were identified using the IQR method.

CONCLUSION:

The sales and customer datasets were successfully combined and analyzed using Python. Correlation analysis was used to study relationships between customer age, purchase frequency, discount, and purchase amount. Customers were grouped into meaningful age categories, while the IQR method was used to identify unusual purchasing patterns.

The visualizations provided a clear understanding of customer purchasing behavior and relationships between different variables. Overall, the analysis demonstrates how data analytics can help businesses understand customers, identify purchasing patterns, segment customers, and make better data-driven marketing decisions.